

Wait... let's Backup

Level 2: Step 1
HARD

Theme: Disconnected Systems: Disks and Storage

Due Date: 6/11/2022

Please take the following notes in consideration:

- In this assignment the questions marked with a (*) are practical and have to be shown to the teacher (no report is needed).
- The remaining questions are to be answered in a written report, to be delivered to the teacher.
- Whenever possible, the VMs necessary to resolve the practical questions should be created as *linked clones*.

Disks, Partitions & File Systems

1. Describe the ZFS, including the advantages and disadvantages over other file systems and the current state of support in Linux.
2. Investigate the purpose and benefits of a **Hot-spare** in the context of RAID storage. In which RAID levels does it make sense?
3. (*) Install a Windows VM, ensuring a 32GB system disk. Install a new instance of Windows in a 16GB VHDx (*Virtual Hard Drive*) disk, created in the Windows VM. At last, verify (and demonstrate) that it is possible to boot, from the boot menu, either Windows instance - the VM and the one in the VHDx.

RAID & Volume Management

4. (*) Create a **Debian Linux VM** with **6 virtual disks** of 100MB each plus the system disk. Describe the necessary steps to perform the following sequence of tasks in the command line:
 - (a) Using **mdadm**, organize the 6 disks in a **RAID 6** array with 4 active and 2 *hot-spare*. Name your device `/dev/md0`.
 - (b) Confirm the layout of your array by inspecting the details of the `/dev/md0` array.
 - (c) Add 2 additional 200MB disks to the VM.
 - (d) Using **LVM (v2)** create **physical volumes** on `/dev/md0` as well as the 2 additional disks.
 - (e) Create a **group of LVM volumes**, named `vg0` containing the physical volumes created for `/dev/md0` and **the first additional disk**.
 - (f) Create a linear LVM **logical volume** named `lv0` with a size of 200MB using the `vg0` volume group. Display the `vg0` volume group information.
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- (g) Create a **JFS file system** in `lv0` and mount it on `/mnt/vg0-lv0`. Ensure the volume is mounted at boot time.
 - (h) Add the second (200MB) additional disk to the `vg0` volume group. Confirm the extra space was added.
 - (i) create a **mirrored logical LVM volume** (1 *mirror*) with the name `lv1` and size 52MB. Confirm that `vg0` free space is now **2×52MB** less.
 - (j) Create a **striped logical LVM volume** (2 *stripes*) with name `lv2` and size 190MB.
 - (k) Increase the size of `lv0` by +60MB. Also resize it's filesystem.
 - (l) Cause the failure of an active disk on `/dev/md0` and remove it from the array. Prove that one of the spare disks took its place by inspecting `/dev/md0` details.
5. Considering the previous scenario, present a list that shows, for each physical volume:
- The used physical extends (PE) and which logical volume are using them
 - The used physical extends
6. (*) Create a **Debian Linux VM** with **5 virtual disks** of 1GB and the system disk. Describe the necessary steps to perform the following sequence of tasks in the command line:
- (a) Enable the `contrib` and `non-free` on package sources and install the Linux Open-ZFS packages (`zfsutils-linux`).
 - (b) Create a **zfs pool** named `userpool` in RAIDZ1 and comprising 3 disks, and a mirrored SLOG in the fourth and fifth.
 - (c) Create a **compressed zfs volume** (`lz4`) with the name `userpool/users`
 - (d) Confirm that `/userpool` and `/userpool/users` were mounted.
 - (e) Compare the IO benchmark with the previous scenario (question 4). You can use the utility `dd`.

Local Backups

- 7. Research and describe what *deduplication* is.
- 8. **Time Machine** application is a very useful tool for macOS users. Describe 3 main features.
- 9. (*) In an **Debian Linux VM** install and demonstrate the *Back In Time* utility.

Deliverables

10. All deliverables submitted via [discord#assignments-submission](#)):
- 1. PDF file with the report
 - 2. The file name should be `12s1h_aXXXXX_aYYYYY.pdf`. This code means:
 - Level 2
 - Step 1

- Hard difficulty
- Students that participated in the group
- **Maximum:** 2 students